Foundation of Special and Inclusive Education

Learning Module No. 03

STUDENT
Name:
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Program:
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LEARNING MODULE INFORMATION	
I. Course Code	
II. Course Title	Foundation of Special and Inclusive Education
III. Module Number	03
IV. Module Title	Student who are Gifted and Talented
V. Overview of the Module	This module covers the central concepts of physical disabilities, health impairments and severe disabilities, giftedness and talent, the theories, and definitions of human intelligence with an expanded presentation on the multiple intelligences' theory by Howard Gardner. The groundwork for a lifetime of intelligence traces the essential concepts on the development of the brain. It also covers the emerging paradigms and various definitions of giftedness and talent, the characteristics of gifted and talented persons, assessment procedures and instructional systems are presented as well.
VI. Module Outcomes	At the end of this module, the students are expected to look for references and materials on great people of the 20th century- the leaders, activists, pioneers, innovators, scientists, and creators. Write a brief paper about each one of them discussing about their exceptionalities and giftedness. Note: One person per category only.
VII. General Instructions	The module is designed to assess student understanding of the assigned lessons found within the associated content of the preliminary term, midterm and final term periods of the course. The assessment part of the module is composed of varied types of questions. You may see true/false, traditional multiple choice, matching, multiple answer, completion, and/or essay. Pay attention to the answer to the assessment questions as you move through each lesson. After each module you will be given a summative test. Your responses to the assessment parts of the module will be checked and recorded. Because the assessment questions are available within the whole completion period and because you can reference the answers to the questions within the content modules, we will not release the answers within modules. However, your professors are happy to discuss the assessments with you during their consultation time, should you have any questions. Good luck. You may not work collaboratively. This is independent work.

Lesson 1: Physical Disabilities, Health Impairments and Severe Disabilities

Lesson Objectives:

1. define, compare, and contrast the terms physical disabilities, health impairments and severe disabilities;
2. describe the skeletal and the muscle systems of the human body;
3. define and differentiate orthopedic from neurological impairments;
4. enumerate and describe the types and classifications of physical disabilities;
5. identify and discuss the chronic illnesses and health-related conditions;
6. enumerate and describe the severe and multiple disabilities; and
7. enumerate and describe the educational programs and support services for students with physical disabilities, healthy impairments and severe disabilities.

Getting Started:

Jerry's Story

Jerry is a 53-year-old father of four children. He's independent, has a house, raised a family and his adult kids still look to him for support. Jerry recently retired as a computer programmer in 2009, and competes and coaches in several sports. This "healthy, everyday Joe, living a normal life" has even participated in the Boston Marathon.

Jerry also has had a disability for over 35 y ears. In 1976 on December 3 (the same day that International Persons with Disabilities Day is recognized) Jerry was hit by a drunk driver. The accident left him as a partial paraplegic.

Jerry's life is not defined by his disability. He lives life just like anyone else without a disability would live their life. "There's lot I can do, and there are some things that I can't do," said Jerry. "I drive, I invest money. I'm not rich, but I'm not poor. I enjoy being healthy, and being independent."

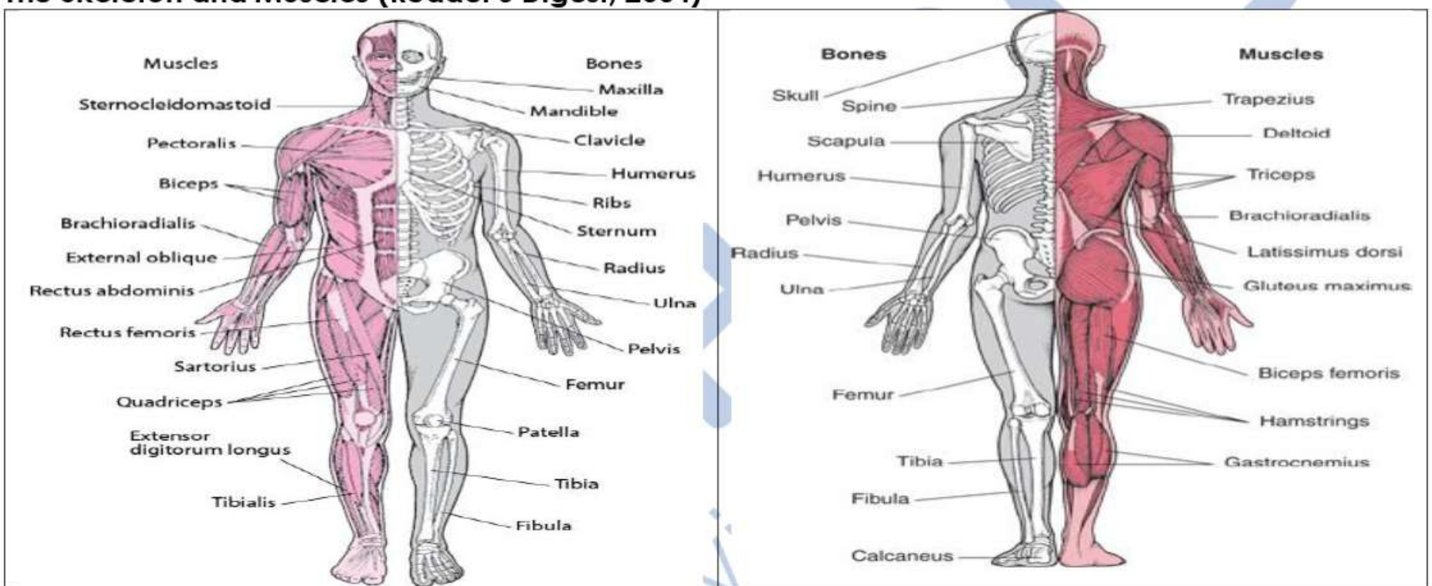
As a person with a disability, however, Jerry has experienced many barriers. Recovering from recent rotator cuff surgery, his rehabilitation specialists "couldn't see past his disability", administering tests and delivering additional rehabilitation visits that a person without a disability wouldn't receive. He once was being prepared for surgery when a nurse proclaimed "he doesn't need an epidural, he's a paraplegic." Jerry had to inform the nurse that he was only a partial paraplegic and that he would indeed need an epidural.

Jerry was in line at an Alabama court house to renew his parking permit and also renew his son's registration. He watched a worker walk down the line and ask people "what do you need?" When she got to Jerry and saw his wheelchair, he was asked "who are you here with?" And Jerry finds it difficult to go to concerts and baseball games with a large family or friends gathering, because rarely are handicap-accessible tickets available for more than two people.

Jerry has seen a lot in over 35 years as someone living with a disability. He's seen many of the barriers and attitudes towards people with disabilities persist. But he's also seen many positive changes to get people with disabilities physically active through recreational opportunities such as golf, fishing and even snow-skiing. There are now organizations such as Lakeshore Foundation external icon – where Jerry works part-time coaching youth basketball and track – that provide recreational opportunities.

Jerry states: "I don't expect the world to revolve around us. I will adapt – just make it so I can adapt.

Discussion:
Human Body's Skeletal and Muscle Systems
The Skeleton and Muscles (Reader's Digest, 2004)



Physical disabilities attack the body's skeletal and muscle systems, the nervous system, the bones, joints and limbs. A brief review of the normal skeletal and muscle systems provides the background for a better understanding of physical disabilities.

The Skeleton

The adult skeleton provides the body's internal scaffolding; it is made up of 206 bones and accounts for one-fifth of the body's total weight. There are four classes of bones:

- Long bones and limbs
- Short bones in limbs and ankles
- Flat bones in the skull
- Irregular bones in the face and vertebrae

The bones of the hands and feet constitute half the total number of bones in the body. The 206 bones are foundin the different parts of the body.

Skull	22	Pertoral Girdle	4
Ears	6	Hip Bones	2
Vertebrae	24	Arms (2x30)	60
Sternum	3	Legs (2x29)	58
Throat	1		

An adult's spine consists of 26 bones called vertebrae. It is divided into four sections;

- Cervical vertebrae – the top seven bones of the spine in the neck
- Thoracic vertebrae – the 12 vertebrae attached to the ribs
- Lumbar vertebrae – the five vertebrae below the ribs
- Sacrum and coccyx – the sacrum is made up of five vertebrae and the coccyx of four. In adults the vertebrae are fused together.

The skeleton supports the body, protects the internal organs and allows a wide variety of movement. Most bones are connected together by ligaments to form flexible joints.

- The skull is formed from 22 different bones. The bones that form the braincase (cranium) are separate at birth but gradually fuse together through childhood.
- The collarbone or clavicle supports the upper arm and allows it to move in a range of direction
- The shoulder blade
- The humerus
- The ribs protect internal organs and the chest cavity.
- The breastbone of sternum is a bony plate that connects the ribs.
- The radius
- The ulna
- There are eight wrist bones or carpals in each wrist.
- The metacarpal bones are the five long bones of the hand that lead to the fingers and thumb.
- The hip or ilium is the outer part of the pelvic girdle.
- The kneecap or patella is the small bone that sits inside the cord-like tendon joining the thigh and muscles.
- The thigh bone or femur is the longest and strongest bone in the body.
- The shinbone or tibia is the major load-bearing bone of the lower leg.
- The fibula is the smaller of the two lower leg bones.

Cartilage is a type of connective tissue that forms shock absorbing discs between vertebrae, gives elasticity and strength to the knee joint, and surrounds the end of every long bone where it meets other bones to form a joint. It also joins the ribs to the breastbone.

Muscles

Muscles control movement throughout the body, including automatic actions such as heartbeat, movement of the gut and blinking. Muscles make up over half of the body's total weight.

There are three kinds of muscles:

Stripe muscle, so-called because of its striated appearance under the microscope, makes up the majority. It contracts in response to messages from the brain.

Smooth muscle is not under conscious control. It controls the digestive, urinary, reproductive and circulatory systems, and such unconscious responses as adjusting the iris in the eye.

Cardiac muscle is found only in the heart, and is unique in being able to contract rhythmically and continuously.

Muscles produce movement by contracting and are arranged in opposing pairs or groups. To raise the forearm, the biceps at the front of the upper arm contract and shorten while the triceps at the back relax and lengthen. To lower the forearm, the actions of these muscles are reversed. The biceps are stronger than the triceps because raising the arm work against the pull of gravity.

- Trapezius is a large, diamond-shaped muscle in the upper back. It holds the head straight and contracts to pull it backwards.
- Shoulder deltoid raises the arms outwards from the body.
- Triceps contract to straighten the arm.
- Pectoral muscle
- Biceps
- External Obliques contract to twist the torso.
- Latissimus dorsi is the large back muscle that holds the body upright.
- Trunk deltoids contract to bend the body forward.
- Thigh muscle or quadriceps is the large muscle that pulls the lower leg forward when walking and holds the leg straight when standing.
- Gluteus maximus is the largest muscle in the body located in the buttocks.
- Hamstring muscles contract to bend the leg and the knee.
- Calf muscle contracts to pull the heel upwards and lift the back of the foot off the ground.
- Achilles tendon is a tough cord that links the bottom of the calf muscle to the heel and pulls the heel upwards when the calf muscle contracts.

Types of Physical Disabilities

Physical disabilities refer to impairments that are temporary or permanent that:

- affect the bones and muscle systems and make mobility and manual dexterity difficult and/or impossible;
- cause deformities and/or absence of body organs and systems necessary for mobility; and affect the nervous system making mobility awkward and uncoordinated

Orthopedic and Neurological Impairments

The two distinct separate types of physical disabilities are **orthopedic** impairments and **neurological** impairments. Joints, limbs, and associated muscles of the skeletal system.

An orthopedic impairment affects the bones, joints, limbs, and associated muscles of the skeletal system. Example of orthopedic impairments are:

1. **Poliomyelitis** also known as infantile paralysis
2. **Osteomyelitis** or tuberculosis of the bones and spine
3. **Bone fracture** or breakage in the continuity of the bone results from falls and accidents
4. **Muscular dystrophy** is a group of long-term diseases that progressively weakens, deteriorates and wastes away the muscles of the body. The disease is usually fatal. The infant appears normal at birth and the muscles begin to weaken at age 2 to 6 years old. The calf muscles appear unusually large because the degenerated muscle has been replaced by fatty tissue. The child walks with an unusual gait, a protruding stomach and hollow back. He or she experiences difficulty in running or climbing stairs, getting to his or her feet after lying down or sitting on the floor and may fall easily.
   By age 10 to 14, the child loses the ability to walk. The small muscles of the hand and fingers are usually the last to weaken and deteriorate.
5. **Osteogenesis imperfecta** is a rare inherited condition marked by extremely brittle bones. The skeletal system does not grow normally, and the bones are easily fractured. As the child matures, the bones may become less brittle, and require less attention.
   The child with osteogenesis imperfecta is fragile and maybe able to walk for short distances with the aid of braces, crutches and other protective equipment.
6. **Limb deficiency** refers to the absence or partial loss of an arm or leg. The Greek word" plegia" which means "to strike" is used in combination with the affected limb, that is, arm or leg, to describe the condition.
   - *Quadriplegia.* All four (quadri) limbs, both arms and legs, are affected. Movement of the trunk and face may also be impaired.
   - *Paraplegia.* Motor impairment of the legs only.
   - *Hemiplegia.* Only one (hemi) side of the body is affected, for example, the left arm and the left leg may be impaired.
   - *Diplegia.* Major involvement of the legs, with less severe involvement of the arms.
   - *Monoplegia.* Only one (mono) limb is affected.
   - *Triplegia.* Three limbs are affected.
   - *Double hemiplegia.* Major involvement of the arms, with less severe involvement of the legs.
   - Orthopedic and neurological impairments have close relationship. For example, a child who is unable to move his legs due to damage in the central nervous system may also develop disorders in the bones and muscles.

An infant may be born with a congenital limb deficiency. Acquired limb deficiency results from the amputation of the arms or legs due to diseases or accidents that require surgery. A prosthesis or artificial limb is often used to replace the limbs to facilitate balance and to create a more normal appearance. Braces, crutches and wheelchairs are used wherever applicable.

7. **Crippling Conditions** that are congenital or present at birth include:

- *Clubfoot* – The child is born with one or both feet deformed usually with the feet and toes turned inward, outward or upward often accompanied by webbed toes.
- *Clubhand* – The same as clubfoot but this time the hands and fingers are deformed.
- *Polydactylism* – The child is born with extra toes or fingers.
- *Syndactylism* – The fingers or toes or both are webbed like those of fowls, ducks and hens.

A neurological impairment involves the nervous system and affects the ability to move, use, feel, or control certain parts of the body. Some of the types of neurological impairment are:

Cerebral Palsy

Cerebral palsy (CP) is a group of disorders that affect a person's ability to move and maintain balance and posture. CP is the most common motor disability in childhood. *Cerebral* means having to do with the brain. *Palsy* means weakness or problems with using the muscles. CP is caused by abnormal brain development or damage to the developing brain that affects a person's ability to control his or her muscles.

The symptoms of CP vary from person to person. A person with severe CP might need to use special equipment to be able to walk, or might not be able to walk at all and might need lifelong care.

A person with mild CP, on the other hand, might walk a little awkwardly, but might not need any special help. CP does not get worse over time, though the exact symptoms can change over a person's lifetime.

All people with CP have problems with movement and posture. Many also have related conditions such as intellectual disability; seizures; problems with vision, hearing, or speech; changes in the spine (such as scoliosis); or joint problems (such as contractures).

Specific Types of Cerebral Palsy
Hypertonia VS Hypotonia

Hypertonia is a lifelong condition where there is too much muscle tone, meaning that limbs are stiff and difficult to move. Hypertonia affects how easily a joint can move, and if for example this affects the legs then it can lead to stiffness, which can lead to falls. In very severe cases it can lead to a frozen joint, which is known as contracture.

Hypotonia symptoms are the opposite of those seen in hypertonia, and muscles will appear to be loose and floppy. Those with hypotonia may have difficulty in maintaining a good posture, may struggle to walk or stand unaided and may also have respiratory problems. Hypotonia is usually caused by damage to the brain suffered within the womb, as well as problematic deliveries.

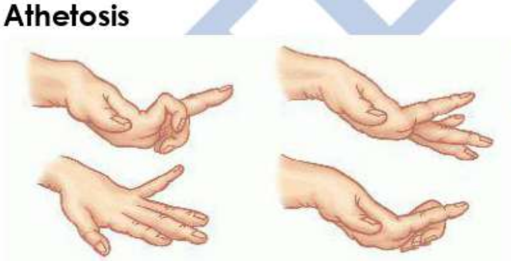
Treatment for hypertonia includes the prescription of muscle relaxants, exercise and physiotherapy. Physiotherapy aims to maintain and increase range of movement, and also aims to avoid contracture. The success of treatment will depend on the severity of the hypertonia, and the level of pain experienced by the sufferer. There are also surgical interventions available to restore movement to limbs by relieving tight muscles and tissues. A relatively new surgical option is SDR (selective dorsal rhizotomy) which aims to reduce stiffness in the legs.

Treatment for hypotonia includes physiotherapy which should be started as soon as possible following diagnosis. A full evaluation of abilities should be conducted to be sure that the most appropriate activities and therapy is provided.

For both hypotonia and hypertonia there are a range of aids available to assist with mobility, such as leg and arm braces which aim to support the limbs.

Occupational therapy aims to identify problems which the sufferer is experiencing with everyday tasks and can advise the best way to carry out complex activities such as getting dressed. This can be invaluable for boosting self-esteem and independence. Occupational therapists can also advise adults with cerebral palsy on independent living.

Athetosis



Athetosis is a movement dysfunction. It's characterized by involuntary writhing movements. These movements may be continuous, slow, and rolling. They may also make maintaining a symmetrical and stable posture difficult.

With athetosis, the same regions of the body are repeatedly affected. These typically include the hands, arms, and feet. The neck, face, tongue, and trunk can be involved, too.

While athetosis may be continuous, it can get worse with attempts to control movement. For example, if a person with the condition tries to type on a computer keyboard, they may have extreme difficulty controlling where their fingers land and how long they remain.
Learning about the symptoms of athetosis and what causes it can help you better understand if the condition is affecting you or someone you love.

Signs and symptoms of athetosis include:

- slow, involuntary, writhing muscle movements
- random and unpredictable changes in muscle movement
- worsening symptoms with attempts at controlled movement
- worsening symptoms with attempts at improved posture
- inability to stand
- difficulty talking

People with athetosis may also experience muscle "overflow." This occurs when you attempt to control one muscle or muscle group and experience uncontrolled movement in another muscle group. For example, when you attempt to talk, you may see increased muscle activity in the arm.

Ataxia

Ataxia is a degenerative disease of the nervous system. Many symptoms of **Ataxia** mimic those of being drunk, such as slurred speech, stumbling, falling, and incoordination. These symptoms are caused by damage to the cerebellum, the part of the brain that is responsible for coordinating movement.

Ataxia describes a lack of muscle control or coordination of voluntary movements, such as walking or picking up objects. A sign of an underlying condition, ataxia can affect various movements and create difficulties with speech, eye movement and swallowing.
Persistent ataxia usually results from damage to the part of your brain that controls muscle coordination (cerebellum). Many conditions can cause ataxia, including alcohol misuse, certain medication, stroke, tumor, cerebral palsy, brain degeneration and multiple sclerosis. Inherited defective genes also can cause the condition.
Treatment for ataxia depends on the cause. Adaptive devices, such as walkers or canes, might help you maintain your independence. Physical therapy, occupational therapy, speech therapy and regular aerobic exercise also might help.

Rigidity

Rigidity is characterized by the markrd resistance of the muscles to passive motion and display extreme stiffness in the affected limbs.

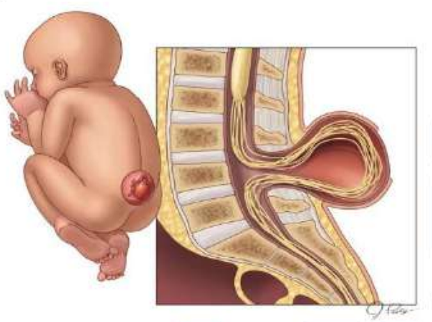
Tremor

Tremor is marked by rhythmic, uncontrollable movements or trembling of the body or limbs.

Mixed Type

This is characterized by the presence of traits mentioned in the preceding categories.

Spina Bifida



Spina Bifida is the most common permanently disabling birth defect that is associated with life in the United States. It's a type of neural tube defect (NTD) that occurs when a baby's neural tube fails to develop or close properly - the literal meaning for Spina Bifida is "split spine". Typically occurring within the first 28 days of pregnancy, while the neural tube is forming, Spina Bifida often occurs before a woman knows she is pregnant.

Spinal Cord Injuries

Spinal cord injuries are results of accidents. Injury to the spinal column is generally described by letters and numbers indicating the site of the damage. For example, C5-6 means the damage has occurred at the level of the 5th and the 6th cervical vertebrae, a flexible area of the neck susceptible to injury from whiplash and driving.

In general, paralysis and loss of sensation occur below the level of injury . The higher the injury on the spine and the more injury (lesion) cut through the entire cord,the greater the paralysis.

Traumatic Brain Injury

Traumatic brain injury is commonly caused by injuries to the head results from automobile, motorcycle, anf bicycle accidents, falls, assaults, gushot wood and child abuse. Severe head trauma offencauses comma or an bnormal deep stupor from which it may be impossible to arouse the affected individual by external stimuli for an extended period.

Temporary or lasting symptoms may include cognitive and language deficits, memory loss, seizures and perceptual disorders. Children may also have difficulty in paying attention and may display inappropriate or exaggerated behavior ranging from extreme aggressiveness to apathy.

Health Impairments

Health impairments of children are due to chronic or acute health problems that adversely affect their educational performance. These problems are present over long periods and tend to get better or disappear. Children with these health problems are not usually confined in hospitals except during attacks or flare ups of their diseases. In many instances, health impairments could result to poor school performance and social acceptance and their effects to the child is great; hence, it is important for teachers to be aware of these problems.

Among the chronic illnesses and other healthy-related conditions are the following:

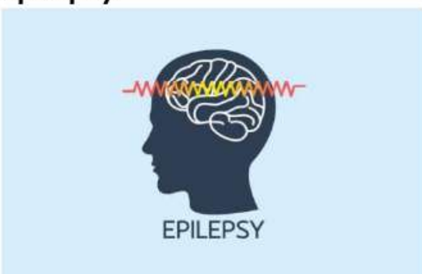
Asthma

Asthma is a chronic lung disease characterized by episodic bouts of wheezing, coughing, and difficulty in breathing due to the inflammation of the airways in the lungs. It is usually triggered by allergens (certain foods, pets, pollen), irritants (smog, cigarette smoke), exercise or emotional stress.

Diabetes

Diabetes is a disorder of metabolism that affects the way the body absorbs and breakdown sugars and starches in food. Children with diabetes have insufficient insulin, a hormone normally produced by the pancreas necessary for proper metabolism and digestion of food.

Epilepsy



Epilepsy is a convulsive disorder commonly known as seizure, a disturbance of movement, sensation, behavior and/or consciousness caused by abnormal electrical activity in the brain. The specific cause of epilepsy is not clearly known.

Hemophilia

Hemophilia is a rare hereditary disorder in which the blood does not clot as quickly as it should. The most serious consequences are usually internal, rather than external contrary to popular opinion. Internal bleeding can cause swelling, pain, and permanent damage to joints, tissues and internal organs; may necessitate hospitalization for blood transfusion.

Burns

Burns result from household accidents but sometimes caused by child abuse. Children with serious burns usually experience pain, scarring, limitations of motion, lengthy hospitalization and repeated surgery. Serious burns can cause complications in other organs.

Severe and Multiple Disabilities

This category includes children who exhibit two or more disabilities, except for the deaf-blind. Usually the combinations are mental retardation, learning disabilities, autism, down syndrome, speech disorders, physical disabilities, visual and hearing impairments, emotional and behavioral disorders, and others. However, while they need intensive support for life, there are those who have average or above average intellectual abilities, giftedness, and talent.

These children have a wide range of characteristics brought about by the combination and severity of the disabilities and age of onset. Some traits are shared like limited speech communication, difficulty in basic physical mobility, problems in generalizing skills from one situation to another, restrictions in relating or attending to others, tendency to forget skills through disuse, and need for intensive and pervasive support in major life activities.

Characteristics

The disabilities significantly affect the development of cognitive, social, emotional and adaptive skills. Learning does not take place the way it does with most children. The condition limits interaction with other children, adults, and restricts normal activities. They have difficulties with adaptive behavior and in coping with the natural and social demands of the environment.

They may display age-inappropriate and socially unaccepted behaviors brought about by the inability to recognize situations when and where certain acts maybe improper and permissible.

Examples are certain forms of antisocial behavior such as undressing in public, self-stimulatory behavior like touching parts of the body, self-injurious behavior like head-banging, poking the eyes and self-mutilation.

They may appear to be completely out of touch with reality and may not show normal human emotions. It may take some work and effort to capture the child's attention or to evoke any observable response.

Many of these children need intensive and pervasive support under constant supervision since they are often unable to care for their basic need as, such as dressing, eating, toileting and maintaining personal hygiene.

Educational Programs

Educational programs and services to children with physical disabilities, health impairments and severe disabilities take into consideration both administrative and instructional models of special education.

Administration of Special Education

Four administrative models are appropriate for this group of children:

1. Inclusion in the regular class. This setting advocates for inclusive education mainly through mainstreaming and full-time placement in regular classes. In this model, the special education teacher helps the regular teacher in dealing with the special needs of children. Ideally, there should be access to the services of an interdisciplinary team composed of physical therapists, occupational therapists, speech therapists, physicians, nurses and other specialists, psychologists and guidance counselors. The amount of support that may be required by these children varies greatly according to the condition, needs and level of functioning.

Parents of those children with chronic illnesses who are often absent from school be assisted to tutor their children at home if they cannot hire special tutors. This way, the children will be able to cope with the lessons discussed in class.

An effective inclusive education program develops self-direction and independence as the students with disabilities attend to their schoolwork. Peer relationships and social development are enhanced by the cooperative class activities that allow the nondisabled students to recognize the abilities of their classmates with handicapping conditions. Research data show that, with adequate support services and class modifications made available, attending the same school and participating in the same activities with their nondisabled classmates greatly contribute to the social development of these children.

2. Special Class. The special is composed of children with disabilities who do not meet the criteria for inclusion in the regular class. While the special education teacher handles the class, partial mainstreaming in regular classes may be worked out. There may be more than one grade level in a special class. The class may be located in the Special Education Center or special education resource room in the regular schools or in special schools.

3. Hospital-bound instruction. The special education program of the hospital admits children with physical disabilities or chronic illnesses who cannot study in regular schools. The national Orthopedic Hospital (NOH) located in Quezon City has a School for Crippled children that offers educational services from elementary to secondary level. The Philippine general Hospital in Manila has a ward known as "Silahis ng Kalusugan" that caters to children who are suffering from chronic illnesses and diseases.

4. Homebound or home-based instruction. Children who have severe or multiple disabilities, mobility problems, or chronic illnesses are regularly visited but itinerant special education teachers in their home who provide instruction based on their needs and capabilities. In case a SPED teacher is not available, the parents can provide direct instruction or hire a teacher to provide regular instruction. The knowledge and skills learned at home or in the hospital can be accredited through the Philippine Educational Placement Test (PEPT) of the Department of Education.

Instructional Models

Modifications and adaptations in the curriculum and strategies in teaching evaluation of learning are introduced to suit the needs and conditions of children with physical disabilities, health impairments severe disabilities.

1. **Individualized Instruction Model.** The Individualized Education Plan or IEP identifies the annual goals, short-term objectives and weekly or daily instructional plans for specific children. Self-management skills are taught for self-regulation and competence in the adaptive skills. A structured curriculum is used and the academic subjects are learned through diagnostic and prescriptive teaching, task analysis, monitoring of progress, and reinforcement through immediate feedback mechanism.
2. **Resource Room Model.** Children who are mainstreamed in regular classes go to the resource room for special instruction, tutorial and mentoring activities. The resource room is the repository for instructional materials and references that the children use in doing their homework and projects to comply with the requirements in the regular class. The resource room is used for meetings with regular teachers, administrators, parents and others. Non-disabled classmates go

to the resource room to do work with the children with disabilities or simply to socialize with each other.

3. **Curriculum Modification Model.** Changes are introduced in the regular curriculum to accommodate the special education needs of the children in a functional curriculum. The functional curriculum includes training on gross and fine motor skills, maximum use of vision, hearing, touch, and other working sensory modalities, receptive and expressive communication skills, functional academics, and social skills. Active rather than passive learning is employed.
4. **Instructional Variations Model.** A variety of suitable and effective instructional approaches are employed. Choices of instructional strategies are based on successful previous preferences of the students, motivation level, individual learning style, and learning rate.

Educational Support Services

Students with physical disabilities, health impairments and severe disabilities need the services of an interdisciplinary team composed of the special and regular teachers and school administrators, parents, physical and occupational therapists, psychologists and guidance counsellors. The team addresses the medical, educational, therapeutic, vocational, and social needs of the children. The job descriptions of the interdisciplinary team members are presented below.

The **physical therapist** (PT) is primarily concerned with the planning and implementation of the program to development and maintains correct bodily posture and mobility. The program objective is to assist the child in the use of muscles and locomotor functions to reduce pain, discomfort or long-term physical damage. Ultimately, efficiency and independence in the use of motor skills and the development of muscular functions is achieved. Message and prescriptive exercises are used together with activities on proper sitting position, special positioning for toileting and feeding, use of special equipment, and play programs, and swimming that children with or without disabilities can enjoy together.

The **occupational therapist** (OT) focuses on the child's participation in activities that are useful in self-care, communication, recreation, employment and other daily living skills. Specific training is given to motor behaviors like drinking from a modified cup, buttoning clothes, tying shoes, pouring liquids and other adaptive skills. These activities enhance the child's physical development, independence, vocational potential and self-concept.

The **speech therapist** (ST) deals with the remediation of all forms of speech, voice, hearing and language problems caused by physical, mental or psychological disorders.

Physicians, nurses and specialists appraise the current health status and the disability itself, provide treatment and recommend therapy services when needed. The **prosthetist** designs and fits artificial limbs; the **orthotist** designs and fits braces and other assistive devices; the **biomedical engineer** develops or adapts assistive technology; and the **social** worker assists students and families in their adjustment to disabilities. The **psychologist** conducts psychodiagnostics and educational assessment for use as basis in school program planning and intervention. The guidance counselor offers individual counselling, guidance service and family therapy to those who need these services.

Application:

Find out how much you learned in this lesson, by answering the question below.

1. Describe the skeletal and muscle systems of the human body. What is the importance of keeping the bones and muscles healthy?

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems overdone.	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

Adapted from: http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf

Summary of the Lesson:

Physical and health impairments are considered disabilities when they interfere with education and other daily living. Physical impairments include those caused by congenital abnormality impairments caused by disease and impairments from other causes. Health impairments include disease and/or chronic illness.

With so many different conditions under the category of Physical and health impairments, there are also lots of diverse causes. Cerebral Palsy and Spina Bifida can be caused by abnormal brain growth. Some other disorders are caused genetically, the result of an injury, poisoning, hormonal abnormalities, while some are still unknown.

On the other hand, a growing number of students in our schools are classified as having severe disabilities. The term severe disabilities refer to a deficit in one or more areas of functioning that significantly limits an individual's performance of major life activities.

This definition of severe disabilities is rather vague, but it reflects the official definition used by school systems. In reality, students classified as having severe disabilities display a wide range of abilities and needs that a school team must address. Severe disabilities stem from a vast number of causes—what's most important is their impact on the students. Students with severe disabilities possess a range of potential in the school setting, and it's crucial that their education be individualized.

Assessment:

1. Which of the following accommodations may be specific to students with physical disabilities?
   a. Oral testing rather than written testing
   b. Physical access to the testing site
   c. Frequent breaks during the test
   d. Additional time to take the test
2. The most common educational placement for students with physical disabilities is:
   a. Regular class
   b. Resource room
   c. Home school
   d. Separate setting

3. Which of these is NOT an accommodation used in the classroom to aid students with physical disabilities?
   a. giving enough space for the student to move around the classroom
   b. provide a structured, predictable classroom environment
   c. having the student leave the class to go to needed therapy sessions
   d. allowing the student to use assistive technology, like choice boards
4. Which of these is NOT a challenge for students with physical disabilities?
   a. getting around the classroom safely
   b. not being able to participate in some activities
   c. not being able to hear what the teacher is saying
   d. using certain learning tools and materials
5. What does it mean when a person is said to have a "congenital disability"?
   a. The disability developed after birth.
   b. The disability is undiagnosed.
   c. The person has a disability but doesn't realize it.
   d. The person has had the disability since birth.
6. It is always obvious, if someone has a disability.
   a. True b. False

7. Most people with disabilities cannot work.
   a. True b. False
8. Impairment of normal physical functioning is a(n):
   a. health impairment c. prosthetic
   b. physical disability d. accommodation
9. An illness, disease or condition that requires special care or attention is a(n):
   a. health impairment c. prosthetic
   b. physical disability d. accommodation
10. Discrimination against persons with disabilities is prohibited in ....
    a. the workplace
    b. education
    c. public programs
    d. all of the above

Enrichment Activity:

Think of a person you know who has physical disability. Ask him/her how he/she adjusts to his/her disability. You can also ask him/her about his/her future plans.

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems overdone.	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

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References/Attributions:

- Inciong, Teresita G., Quijano, Yolanda S. and Capulong, Yolanda T. Introduction to Special Education. Manila, Philippines: Rex Bookstore, 2007.

Lesson 2: Theories and Concepts of Intelligence

According to research the prominent men and women from different countries all over the world who have carved a name for themselves in their respective fields of endeavor as well as the many other people who have excelled in their lines of expertise have four things in common: a.) they possess intelligence or high intellectual ability, b.) creativity, c.) talent, and d.) task commitment.

Can you imagine what it is to be like Lea Salonga or Cecille Licad who attained international fame and brought honors to our country through their outstanding achievements in the performing arts at a very young age?

How about our national hero, Dr. Jose Rizal, who is one among the few geniuses of renown in the world?

Lesson Objectives:

1. Discuss the nature of the human intellect as expounded by philosophers, psychologists and educators through centuries;
2. Enumerate and describe the theories and definitions of intelligence; and
3. Enumerate and discuss the multiple intelligences of a person.

Getting Started (Optional):

Meet the gifted

By Nathalie Tomada, The Philippine Star, May 19, 2003

Conversations have never been this interesting. Emil Justin Cebrian talks about his admiration for the wisdom of Confucius, his thoughts on the spread of the SARS epidemic, and his disapproval on the use of contraceptives- just like any learned, opinionated adult. Except that he is only 12 years old.

John Bryan Santiago Tiosin, whose first words when he was about four months old is supercalifragilis... (go figure!) has been ploughing through books at an age when others were just getting past thumb-sucking. From middle earth and Tolkien, he claims to be now smitten with Michaelangelo and the History of Art. Bryan is only seven years old.

Meet the gifted children. "Alam ko naman, higher level ang pag-iisip ko kaysa iba," Justin says, insisting that "most of the time, I don't think about it. I'm really just an ordinary kid." Hardly. According to parents, Fred and Ceres, Emil Justin, named after the great French sociologist Emil Durkhheim, was already talking before he turned one. He mastered the National Anthem, flags, capitals and Philippine presidents before he turned two. After several accelerations, the award-winning storyteller of Museo Pambata is now an incoming senior at Arellano

High School and, as usual, gunning for the highest honors. When that happens, he will perhaps be the youngest valedictorian in the country.

1. What do you think are the characteristics of Justin and Bryan that make them different from other children and youth of the same chronological age?
2. What philosophical thoughts and scientific theories do you think are the reasons behind the human intellect?

Discussion:

The nature of human intellect has fascinated scholars and became the subject of debates, studies and propositions as early as during the time of the Greek philosopher Plato and Aristotle. When the field of psychology began to emerge in the 17th and 18th Centuries as a discipline separate from philosophy, mathematics and biology, individuals such as John Locked, Charles Darwin, Francis Galton and Charcot continued to influence the study of intelligence. A number of prominent European schools of psychology flourished until early part of the 19th century. Thus, as the students of the Great Schools began to form their own programs, the number of theoretical and empirical investigations of intelligence increases.

In the latter part of the 20th century, new statistical designs and modern experimental strategies were developed that made psychological testing popular in most western countries. The theory of multiple intelligences began to appear, particularly the work of Thurstone and Guilford. The prominent theorists were Burt, Thurstone, P. Cattell, Wechsler, Guilford, Vernon, Hunt, Anna Anastasi, Thorndike, Inhelder, Taylor and Eysenck.

Current trends in intelligence theory and research involve the formation of more complex multiple intelligence theories. Standardized tests to measure intelligence are used only as one of the sources of data about mental ability. The fields of genetics and neurological research methodologies on the measurement of intelligence has generated a number of factors on intelligence. In addition to mental ability, other data are considered simultaneously in determining the intelligence level of a person. Data are derived from the environment, biological factors and psychological aspects of the intellect.

Theories and Definitions of Intelligence

1. The Binet-Simon Scale (1890s)

The modern approach to understand the concept of intelligence began with the work of Alfred Binet, a French psychologist (1857-1911) and his colleague, Theodore Simon (1873-1961). Together they developed and co-authored a test to roughly measure the intellectual development of young children between the ages of three to twelve. They wanted to find a way to measure the ability of children to think and reason.

From Binet's work, that term "intelligence quotient" or IQ evolved.

IQ is the ratio of "mental age" to chronological age with 100 as the average.

Example: An 8-year-old who passes the test for an 8-year-olds has an IQ of 100 which is the average for his or her chronological age. While an 8-year-old who passes the test for 10-year-olds has an IQ of 10/8 x 100 or 125. This child's IQ is above the average for his or her chronological age. Thus, he or she is brighter or more superior than the other children his or her age.

2. Spearman's Two –Factor Theory of Intelligence (1904)

Charles Spearman, a British psychologist (1863-1945), advanced the two-factor theory of intelligence "g" and "s".

- the performance of any intellectual act requires some combination of "g" or general factor which is available to the same individual to the same degree for all intellectual acts, and of "s" or specific factors which are specific to that act and which varies in strength from one act to another.
- The theory explains that if one knows how a person performs on one task that is highly saturated with "g", one can safely predict similar level of performance for another highly "g" saturated task. Prediction of performance on tasks with high "s" factors is less accurate. Nevertheless, since "g" pervades all tasks, prediction will be significantly better than chance. Thus, the most important information to have about a person's intellectual ability is an estimate of his "g".

3. Terman's Stanford Binet Individual Intelligence Test (1906)

Lewis Madison Terman, an American cognitive psychologist (1877- 1956), published a revised and perfected Binet-Simon Scale for American populations in 1906 while he was at Stanford University.

- In 1916 he adopted William Stern's suggestion that the ratio between mental and chronological age be taken as a unitary measure of intelligence multiplied by 100 to get rid of the decimals.
- By far, it is the it is considered as the best available individual test of intelligence.

4. Thorndike's Stimulus Response Theory (1920's)

Edward Thorndike, an American psychologist (1874-1949), and his students used objective measurements of intelligence on human subjects as early as 1903. During the 1920s he developed a multifactored test of intelligence that consisted of completion, arithmetic, vocabulary and directions tests (CAVD).

- He drew an important distinction among **three broad classes of intellectual functioning**:
  a. abstract intelligence- measured by standard intelligence tests
  b. mechanical intelligence- the ability to visualize relationships among objects and understand how the physical world works
  c. social intelligence- ability to function successfully in interpersonal situations

Four dimensions of abstract intelligence as proposed by Thorndike:

a. altitude of the complexity of difficulty of tasks one can perform
b. width or the variety of tasks of a given difficulty
c. area which is the function of width and altitude
d. speed which is the number of tasks one can complete in a given time

- He is cited for his work on what he considered as the two most basic intelligences: trial and error and stimulus response association. His proposition stated that stimulus response connections that are repeated are strengthened while those that are not used are weakened.

5. L.L. Thurstone's Multiple Factors Theory of Intelligence (1938)

Louis L. Thurstone was an American psychometrician (1887-1955) who studied intelligence tests and tests of perception through factor analysis. His theory stated that intelligence is made up of several primary mental abilities rather than a general factor and several specific factors.

- **Seven primary mental abilities identified in Thurstone's Multiple Factors Theory of Intelligence:**
  a. verbal comprehension
  b. word fluency
  c. number facility
  d. spatial visualization
  e. associative memory
  f. perceptual speed
  g. reasoning
- He was among the first to propose and demonstrate that there are numerous ways in which a person can be intelligent. His multiple factors theory has been used in the development of intelligence tests that yield a profile of the person's performance in each of the seven primary mental abilities.

6. Cattell's Theory on Fluid and Crystallized Intelligence

Raymond B. Cattell, a British-American psychologist (1905-1998) theorized that there are **two types of intelligence** namely:

a. Fluid intelligence- is essentially nonverbal and relatively culture free. It involves adaptive and new learning capabilities, related to mental operations and processes on capacity, decay, selection and storage of information. It is more dependent on the physiological structures or parts of the brain that are responsible for intellectual behavior. Also, it increases until adolescence, then goes through a plateau and begins to gradually decline with the degeneration of the brain's physiological structures.

b. Crystallized intelligence develops through the exercise of fluid intelligence. It is the product of the acquisition of knowledge and skills that are strongly dependent upon exposure to culture. It is related to mental products and achievement and highly influenced by formal and informal educational factors throughout the life span. Also, it continues to increase through middle adulthood.

7. Guilford's Theory on the Structure of the Intellect (1967)

J.P. Guilford, an American psychologist, advanced a general theory of human intelligence whose major application or use is for educational research, personnel selection and placement and the education of gifted and talented children.

- The theory on the structure of the intellect (SOI)advances that human intelligence is composed of 180 separate mental abilities (the initial count was 120) that have been identified through factor analysis.
- The mental abilities are composites three separate dimensions: contents, operations, and products.

Four types of contents:

1. **figural-** the properties of stimuli experienced through the senses- visual, auditory, olfactory, gustatory, and kinesthetic. Examples: shapes and forms, sizes, colors, sounds, tastes, temperature, intensity, volume.
2. **symbolic-** letters, numbers, symbols, designs
3. **semantic**-words and ideas
4. **behavioral-** actions and expressions of thoughts and ideas

Five kinds of operations:
1. **cognition**- the ability to gain, recognize, and discover knowledge
2. **memory**- the ability to retain, store, retrieve and recall the contents of thoughts
3. **divergent production**- the ability to produce a variety of ideas or solutions to a problem
4. **convergent production**- the ability to produce a single best solution to a problem
5. **evaluation**- the ability to render judgment and decide whether the intellectual contents are correct or wrong, good or bad.

Six kinds of products:
1. units that come in single number, letter or word
2. classes or a higher order concept, for example, men and women= people
3. relation or connection between and among classes and concepts
4. systems or the process of ordering or classification of relation
5. transformation the process of altering or restructuring of intellectual content
6. implication or the process making inferences from separate pieces of information.

8. Sternberg's Triarchic Theory of Intelligence (1982)

Robert Sternberg of Yale University theorized that intelligence is a fixed capacity of a person. Hence with higher intellectual capabilities, like in the case of children and youth who are gifted and talented, almost every task can be achieved at a high level of performance.

- The triarchic theory of intelligence seeks to explain in an integrative way the relationship between:

1. intelligence and the internal world of the individual, or the mental mechanisms that underlie intelligent behavior

2. intelligence and external world of the individual, or the use of these mental mechanisms in everyday life in order to attain an intelligent fit to the environment

3. intelligence and experience, or the mediating role of one's passage through life between the internal and external worlds of individual.

Three main parts or dimensions of Triarchic Theory:

1. Contextual intelligence- intelligence in its sociocultural contexts.

2. Experiential intelligence- emphasizes insights and the ability to formulate new ideas and combine seemingly unrelated facts or information.

3. Componential intelligence- emphasizes the effectiveness of information processing. Sternberg defines component as the underlying cognitive mechanisms that carry out the adaptive behavior to novel situations.

Two kinds of Components

1. Performance components- used in the actual execution of the tasks. They include encoding, comparing, chunking and triggering actions and speech.

2. Metacomponents- the higher order executive processes used in planning, monitoring and evaluating one's working memory program.

9. Gardner's Theory of Multiple Intelligences (1983)

Howard Gardner is a psychologist and professor at Harvard University Graduate School of Education and director of Project Zero.

- He developed his breakthrough theory of multiple intelligences or MI based on his studies of many people from different walks of life in everyday circumstances and professions.
- He did a massive synthesis of a lot of research including brain research, evolutionary research and genetic research.
- He authored 20 books and hundreds of articles on MI.
- He emphasizes that MI is originally not an educational theory. It is a theory on how the mind is organized and developed.
- He describes the mind as having 7,8,9 or even a dozen different "computers" in the brain that determines whether a person is will be competent or incompetent at everything.

The Multiple Intelligences

1. Linguistic Intelligence

- is the ability to use language to excite, please, convince, simulate or convey information.

-it can be developed through the use of the following activities: reading fiction, literary work, newspapers, magazines, reports, biographies, bibliographies, the Internet; engaging in storytelling, debates, plays, listening to audiotapes, watching films, writing reports, stories, speeches.

- practitioners who have high linguistic intelligence include novelists, poets, journalists, storytellers, actors, orators, comedians, politicians.

2. Logical- Mathematical Intelligence

-is the ability to explore patterns, categories and relationships by manipulating objects or symbols and to experiment in controlled, orderly ways.

- it can be enhanced with the use of the following activities: mazes, puzzles, outlines, matrices, sequences, codes, patterns, logic, analogies, timelines, equations, games, formulas, theorems, calculations, computations, syllogisms, probabilities.

- persons who have high logical-mathematical intelligence are mathematicians, scientists, computer engineers and programmers, doctors, astronomers, inventors, accountants, lawyers, economists, detectives, trivia champions.

3. Bodily-Kinesthetic Intelligence

- refers to the ability to use fine and gross motor skills in sports, the performing arts, or arts and crafts production.

- the following activities develop bodily kinesthetic intelligence: role playing, dramatization, skits, mimes, body language, gestures, facial expressions, dancing, sports, games, experiments, laboratory work.

- those who have high bodily-kinesthetic intelligence are: ballet and folk dancers, choreographers, sculptors, professional athletes, gymnasts, surgeons, calligraphers, jewelers, watchmakers, carpenters, circus performers.

4. Spatial Intelligence

- is the ability to perceive and mentally manipulate a form or object, perceive and create tension, balance and composition in a visual or spatial display.
- some activities that enhance spatial intelligence are: illustrations, constructions, maps, paintings, drawings, mosaics, sketches, cartoons, sculptures, storyboards, videotapes.
- those who have high spatial intelligence are: urban-planners, architects, engineers, surveyors, explorers, navigators, mechanics, curators, map designer, fashion designers, florists, interior designers, visual artists, muralists, photographers, movie directors, set designers, chess players, cartoonists.

5. Musical Intelligence

- is the ability to enjoy, perform or compose a musical piece.
-persons who succeed in the following occupations have high musical intelligence: composers, musicians, conductors, critics, opera artists, singers, rappers, instrument makers and players, sound recording artists.

6. Interpersonal Intelligence

-is the ability to understand and get along with others.
-the type of activities that will develop interpersonal intelligence include group projects and charts, communication, social interaction, dialogs, conversations, debates, arguments, consensus building, group work on murals and mosaics, round robins, games, challenges and sports.
- those who have high interpersonal intelligence are: teachers, social workers, doctors and nurses, anthropologists, counselors, priests/ministers, nuns, entrepreneurs, ombudsman, managers, politicians, salespersons, tour guides.

7. Intrapersonal Intelligence

- is the ability to gain access to and understand one's inner feelings, dreams and ideas.
- the activities that will enhance interpersonal intelligence include insight and intuition building, creative and critical thinking, goal setting, reflection and self-meditation, self-assessment, affirmation, keeping journals, logs and reflectionnaires, "I" statements, discussion, interpretation and creative expression of values, philosophical thoughts and ideas, quotations.

8. Naturalist Intelligence

- is the most recent addition to the original list of seven multiple intelligences. It refers to the person's ability to identify and classify patterns in nature.
- in prehistoric times when people relied on hunting animals and gathering plants, this intelligence was used to sort what animals and plants were edible or not. At present, a person uses his or her naturalist intelligence in the ways he or she relates to the environment.
- a person who has naturalist intelligence abilities is likely to be sensitive to changes in flora and fauna, weather patterns and similar environmental factors.

Application

Using the matrix below, compare, and contrast the theories and definitions of intelligence as advanced by the proponents.

Proponent	Theory of Intelligence	Definition of Intelligence
1. Alfred Binet and Theodore Simon		
2. Charles Spearman		
3. Edward L. Thorndike		
4. Robert Sternberg		
5. Howard Gardner		

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems overdone.	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

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Summary of the Lesson:

There is a long list of philosophical thoughts, scientific theories, inventions and technological advances through the centuries that intelligent and creative human minds evolved in the sciences, the various fields in medicine, mathematics, arts and other areas. The achievement introduced dramatic changes in human lives such as increase in the life span, cure for diseases, more convenient, comfortable and enjoyable life styles and information technology.

Several theories and definitions of intelligence was presented by different proponents like Alfred Binet and Theodore Simon, Charles Spearman, Lewis M. Terman, Edward Thorndike, L.L Thurstone, Raymond Cattell, J.P. Guilford, Robert Sternberg, and Howard Gardner.

While a big number of definitions of intelligence have been published, there seems to be no consensus or agreement on what intelligence actually is. As what Catell (1971) puts in, intelligence is a combination of human traits which includes the capacity for insight into complex relationship, all of the processes involved in abstract thinking and a capacity to acquire new capacity.

Assessment:

I. Identification. Write the name or words that is being described in each statement.

____________________1. Author of a contemporary theory of multiple intelligences consisting of eight separate kinds of intelligence.

____________________2. He theorized that with higher intellectual capabilities almost every task can be achieve at a high level of performance.

____________________3. Having a keen eye for details, enjoys puzzles and riddles are indicators of this intelligence.

____________________4. A person who has this intelligence are sensitive to changes in weather patterns and environmental factors.

____________________5. The indicators of this element of multiple intelligence are evidenced by people who are engage in creative thinking, novel and original ideas.

II. Matching Type. Match Column A with those in Column B. Write the letter of your answer on the space before the number in column A.

A	B
________1. Linguistic Intelligence	a. analyzing, comparing, evaluating
________2. Spatial intelligence	b. applying, using
________3. Bodily-kinesthetic	c. solving math and logic problems
________4. Interpersonal intelligence	d. Balance, strength, endurance
________5. Logical-mathematical	e. reading, comprehension and writing
	f. drawing, doodling, sketching
	g. mediator and counselor

Enrichment Activity:

In 1996, the editors of TIME magazine published a special edition that featured the remarkable characters that influenced the forces and great events of the past 100 years. Titled "Great People of the 20th Century", the book presents the biographies and achievements of the most memorable and unforgettable individuals. These individuals are categorized as follows:

1. The Leaders- diplomats and dictators who shaped the destiny of nations
2. The Activists- the men and women who fought for change from outside the traditional halls of power
3. The Pioneers- the men and women who have dared to explore new fields and break down barriers
4. The Innovators- the gifted few whose visions have changed our lives
5. The Scientists- the researchers whose work has revolutionized human society in the span of only 100 years.
6. The Creators- the artists whose work has shaped and mirrored the century

Make a research about the great people of the 20th Century and make a list of 5 people under each category.

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
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Suggested Links (Optional):

Link 1 Theories of Intelligence

- https://www.youtube.com/watch?v=-hMhpB8ikR8

Link 2 Howard Gardner on Multiple Intelligences

- https://www.youtube.com/watch?v=8N2pnYne0ZA

References/Attributions:

- Inciong, Teresita G., Quijano, Yolanda S. and Capulong, Yolanda T. Introduction to Special Education. Manila, Philippines: Rex Bookstore, 2007.

Lesson 3: Gifted and Talented Students (Definitions, Characteristics, and Curriculum)

This covers the central concepts on giftedness and talent, the theories and definitions of human intelligence with an expanded presentation on the multiple intelligences' theory by Howard Gardner. The groundwork for a lifetime of intelligence traces the essential concepts on the development of the brain, the 'seat' of man's intellectual capacity.

Lesson Objectives:

1. Compare and contrast the various definitions of giftedness and talent;
2. Enumerate and discuss the characteristics of gifted and talented children; and
3. Describe the assessment procedures, curricular programs and instructional systems for gifted and talented students.

Getting Started:

"What makes a child gifted and talented may not always be good grades in school, but a different way of looking at the world and learning." - Chuck Grassley

1. What is can you say about this quote? Do you agree with the author?
2. What do you think are the characteristics of gifted and talented students which make them different from their age peers?

Discussion:

Definitions of Giftedness and Talent

Old concepts	Emerging paradigm
Giftedness is high IQ	Many Types of Giftedness
Trait-based	Qualities-based
Subgroup Elitism	Individual Excellence
Innate, "In There"	Based on Context
Test-Driven	Achievement- driven, "What you do" is gifted
Authoritarian, "You are or are not Gifted"	Collaborative, Determined by Consultation
School-Oriented	Field-and- Domain Oriented
Ethnocentric	Diverse

Through the years, the concept on intellectual giftedness had change as shown in the following figure (Heward, 2003).

Figure 1: Changing Concepts on Intellectual Giftedness

The concept of giftedness varies among different scholars. It also varies from different cultures. Davis, Rimm, and Siegle (2011) noted that there is no single definition of the word gifted that is universally accepted. Some scholars refer to it as having special talents and abilities while others think of it as a state of high intellect or genius.

Tannenbaum (2003) proposed a definition of giftedness in children to denote their potential for becoming critically acclaimed performers or exemplary producers of ideas in spheres of activity that enhance the moral, physical, emotional, social, intellectual or aesthetic life of humanity.

For Gagne (1985), giftedness refers to domains of human abilities, talents, to domains of human accomplishments. Giftedness also involves excellence, rarity, productivity, demonstrability, and value attached to the skills/products of the individual (Sternberg & Zhang, 1995).

Federal or American Government's Definitions

The first federal definition of the gifted and the talented was contained in the 1972 Maryland Report. Gifted and talented children are capable of high performance and demonstrate potential ability in any of the following six areas:

- General intellectual ability
- Specific academic aptitude
- Creative or productive thinking
- Leadership ability
- Ability in the visual or performing arts
- Psychomotor ability

The Gifted and Talented Children's Act of 1978

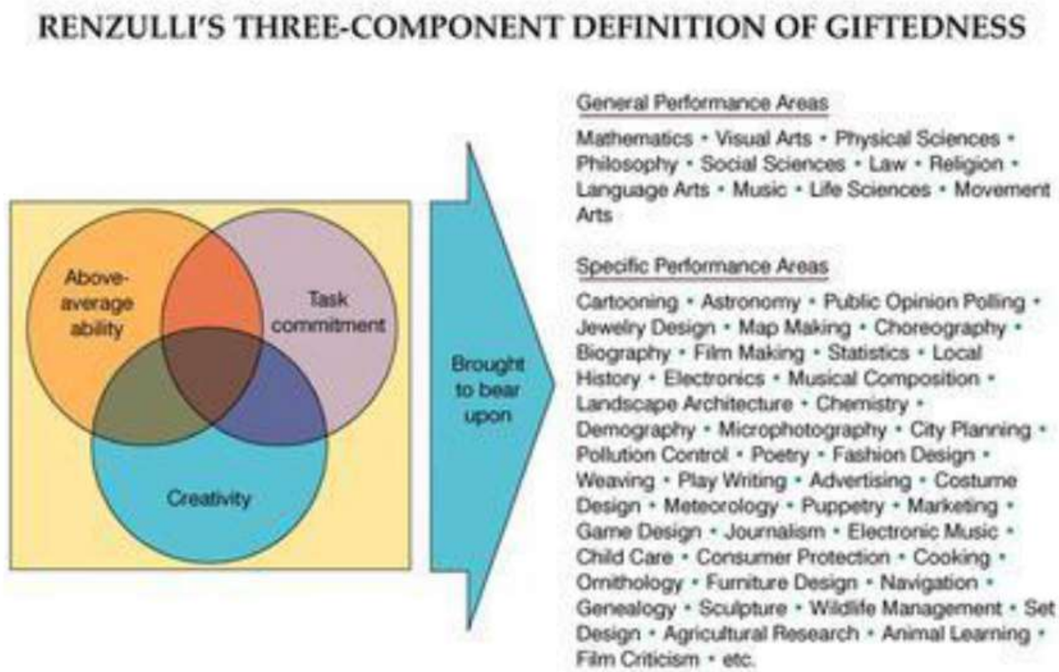
- "gifted and talented children are those who "possesses and demonstrate abilities that give evidence of high-performance capability in such areas as intellectual, creative, specific, academic or leadership ability, or in performing or visual arts and who, by any reason thereof require services or activities not ordinarily provided by the school".

The 1991 "Report on National Excellence: A Case for Developing America's Talent" deleted the term ***gifted*** and used ***outstanding talent*** and ***exceptional talent instead***. This means that talent occurs in all groups across all cultures and is not necessarily revealed in test scores but in a person's "high performance capability" in the intellectual, creative and artistic realms. Thus, Giftedness is said to connote "a mature power rather than a developing ability."

Renzulli's Three-Trait Definition of Giftedness

- the definition states that giftedness results from the interaction of: 1.) above- average general abilities, 2) a high level of task commitment, and 3.) creativity.
- Children who manifest or are capable of developing an interaction among the three clusters require a wide variety of educational opportunities and services that are not ordinarily provided through regular instructional programs. See figure 2.

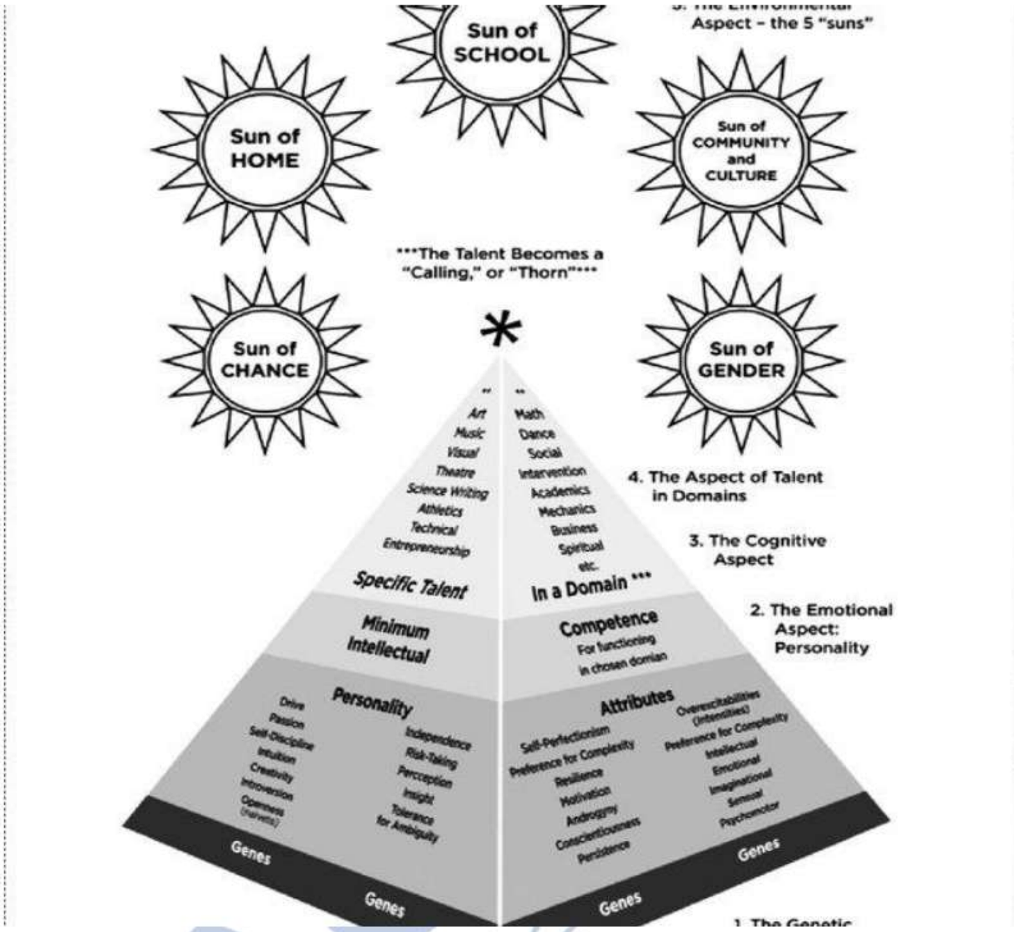
Figure 2:



Piirto's Pyramid Model of Talent Development

- Piirto's 1999 definition states that the gifted are "those individuals who by way of having certain learning characteristics such as superior memory, observational powers, curiosity, creativity, and the ability to learn school-related subject matters rapidly and accurately with a minimum of drill and repetition, have a right to an education that is differentiated according to those characteristics."
- He also states that even if gifted students do not become producers of knowledge or makers of novelty, special education should train them to become adults who will produce knowledge or make new artistic and social products.
- His pyramid model is composed of: 1**.) a foundation of genetic endowment**, 2.) **personality attribute**s such as drive, resilience, intuition, perception, intensity, and the like; 3.) **the minimum intelligence level necessary for function** in the domain in which the talent is demonstrated; 4.) **talent in a specific domain** such as mathematics, writing, visual arts, music, science or athletics; and 5.) **the environmental influences of five suns**: the sun of **home**, **community and culture**, **school**, **chance** and **gender**. See figure 3.

Figure 3



Maker's Problem-Solving Perspective

- Maker in 1996 incorporates high intelligence, high creativity and excellent problem-solving skills.
- He enumerates the following characteristics of a gifted person: a problem solver- one who enjoys the challenge of complexity and persists until the problem is solved in a satisfying way.
- Also, according to him, gifted person is capable of a.) creating a new or more clear definition of an existing problem, b.) devising new and more efficient or effective methods, and c.) reaching solutions that may be different from the usual, but are recognized as being effective than previous solution.

Characteristics of Gifted and Talented Children and Youth

- Giftedness and Talent are a complex condition that covers a wide range of human abilities and traits; thus, it must be clearly understood that they vary according to social context.

- According to Silverman's studies (1995), highly gifted students have IQ scores 3 standard deviations or greater above the mean. The IQ score is greater than 145, or 35 to 55 points more or even higher than the average IQ scores of 90 to 110. Among American children, there is only 1child in 1000 or 1 child in 10,000. Silverman found the following characteristics among these highly gifted individuals:
    - Intense intellectual curiosity
    - Fascination with words and ideas
    - Perfectionism
    - Need for precision
    - Learning in great intuitive leaps
    - Intense need for mental stimulation
    - Difficulty conforming to the thinking of others
    - Early moral and existential concern
    - Tendency toward introversion

Shaklee (1989, cited in Heward, 2003) listed the identifiers of young gifted and talented children as follows:

- Exceptional learner in the acquisition and retention of knowledge:
  a. exceptional memory
  b. learns quickly and easily
  c. advanced understanding/ meaning of arca
- Exceptional user of knowledge in the application and comprehension of knowledge
  a. exceptional use of knowledge
  b. advanced use of symbol systems- expressive and complex
  c. demands a reason for unexplained events
  d. reasons well in problem-solving-draws from previous knowledge and transfers it to other areas
- Exceptional generator of knowledge- individual and creative attributes
  a. highly creative behavior in areas of interest and talent
  b. does not conform to typical ways of thinking, perceiving
  c. enjoys self-expression of ideas, feelings or beliefs
  d. keen sense of humor that reflects advanced, unusual comprehension of relationships and meaning
  e. highly developed curiosity about cause, future, the unknown
- Exceptional motivation- individual motivational attributes
  a. perfectionism: striving to achieve high standards, especially in areas of talent and interest
  b. shows initiative, self-directed
  c. high level of inquiry and reflection
  d. long attention span when motivated
  e. leadership-desire and ability to lead

f. intense desire to know

Think about It!

Which of the following characteristics do you possess? Do you think you are one of the gifted children of your generation?

Creativity as the Highest Expression of Giftedness

Creative ability is considered as central to the definition of giftedness.

a. Clark (1986) refers to ***creativity as the highest expression of giftedness***.
b. Sternberg (1988) suggests that creative, insightful individuals are those who make discoveries and devise the inventions that ultimately change the society.
c. Guilford (1988) enumerates the following **dimensions of creative behavior**:
    - Fluency- the creative person is capable of producing many ideas per unit of time.
    - Flexibility- a wide variety of ideas, unusual ideas, and alternative solutions are offered
    - Novelty/originality- low probability, unique words, and responses are used; the creative person has novel ideas
    - Elaboration- the ability to provide details is evidenced
    - Synthesizing ability – the person has the ability to put unlikely ideas together
    - Analyzing ability – the person has the ability to organize ideas into larger, inclusive patterns. Symbolic structures must often be broken down before they can be reformed into new ones
    - Ability to recognize or redefine existing ideas- the ability to transform an existing object into one of different design, function, or use is evident
    - Complexity- the ability to manipulate many interrelated ideas at the same time is shown

d. Torrance (1993) found in a 30-year longitudinal study that high-ability adults who were judged to have achieve far beyond their peers in creative endeavor possess the following ten most common characteristics:
    1. Delight in deep thinking
    2. Tolerance of mistakes
    3. Love of one's work
    4. Clear purpose
    5. Enjoyment in one's work
    6. Feeling comfortable as minority of one
    7. Being different
    8. Not being well-rounded
    9. A sense of mission
    10. The courage to be creative

Assessment of Gifted and Talented Children

Similar to the screening and location and identification and assessment of exceptional children, the following processes are employed:

1. Pre-referral Intervention

- Exceptional children are identified as early as possible. Teachers are asked to nominate students who may possess the characteristics of giftedness and talent through the use of a Teacher Nomination Form.

2. Multifactored Evaluation

- Information are gathered from a variety of sources using the following materials:
  - Group and individual intelligence tests
  - Performance in the school-based achievement tests
  - Permanent records, performance in previous grades, awards received
  - Portfolios of student work
  - Parent, peer, self-nomination

Curriculum and Instructions for Gifted and Talented Children

Differentiated Curriculum and Instructional Systems

The skills in the Basic Elementary Curriculum of the Department of Education are intended for average learners and lack the competencies that match the learning characteristics of high-ability students. A study of American gifted and talented students found that 60% of all grade four students in a school district have already mastered much of the content of the curriculum. Majority of the students scored 80% in a pretest in mathematics even before the school year began. A ***differentiated*** **curriculum** that is modified in depth and pace is used in special education programs for gifted and talented students.

1. **Curriculum compacting** – is the method of modifying the regular curriculum for certain grade levels by compressing the content and skills that high-ability students are capable of learning in a shorter period of time.
   - At the Silahis Special Education Centers of Manila City Schools, high ability students study the fourth, fifth and sixth grades in a span of two years.
2. **Enrichment** of the regular curriculum allows the students to study the content at a greater depth both in the horizontal and vertical directions employing higher order thinking skills. The differentiated curriculum goes beyond the so-called "basic learning competencies" of BLC and allows the students access to advanced topics of interest to them. Meanwhile, acceleration modifies the pace or length of time at which the students gain the skills and competencies in the regular curriculum to accommodate the enrichment process.
   a. **Horizontal enrichment** adds more content and increases the learning areas not found in the regular curriculum for the grade level. Here the students go beyond the grade requirements and move on to study the subjects in the higher grades.
   b. **Vertical enrichment** allows the students to engage in independent study, experimentation, and investigation of topics that interest them.

Example: Social Studies and Makabayan subjects lend themselves well to vertical enrichment activities that will give high-ability students opportunities to share their ideas in solving related problems at home, the school and the community.

c. **Self-contained Class**- high ability students are enrolled in a special class that is taught by trained special education teacher. Mainstreaming activities are arranged so that the students can socialize with their peers, share their knowledge and assist in peer mentoring the slow learners.

Six Strategies for Challenging Gifted Learners

Dina Brulles, director of gifted education in the Paradise Valley Unified School District in Phoenix, Ariz., believes gifted students need less grade-level work, faster-paced lessons, deeper and more advanced content, and opportunities to work with other gifted students. They also require a different kind of interaction with the teacher, who must be less of a "sage on the stage" and more of a "guide on the side."

With the following strategies, teachers can tend to the complex needs of their high-ability students in the heterogeneous classroom.

1. **Offer the Most Difficult First**
   "Most Difficult First" is one manageable way for teachers to compact the curriculum for their high-ability students. With compacting, students get to "throw away" the part of the curriculum that they already know, while receiving full credit for those competencies. This frees up students to work on more challenging content.
2. **Pre-Test for Volunteers**
   Let's say a teacher is teaching two-digit multiplication. He might do some direct instruction for 10 minutes, then offer students the end-of-chapter test, saying, "If you get 90 percent or higher, you won't have to do the homework or practice work. You'll have different work to do." According to Brulles, some gifted students will take this option, whereas others may decide, "I don't know this; I need the practice work." Again, as in Most Difficult First, this strategy requires having extension work for students who test out of the material.
3. **Prepare to Take It Up**
   Example when the class is working on the distributive property in math, those "piles" might include differentiated worksheets, word problems, and task cards. Depending on how students grasp the concept, the teacher can either reteach, offer practice, or enrich.
4. **Speak to Student Interests**
   According to education expert Jenny Grant Rankin, knowing a student's emotional intensities—what Polish psychologist Kazimierz Dabrowski called "overexcitabilities"—is also key to teaching gifted students. Dabrowski identified five areas of sensitivity that are strongly related to giftedness: psychomotor, sensual, intellectual, imaginational, and emotional. Knowing a student's overexcitabilities can help teachers shape engaging—and personalized—learning experiences.
5. **Enable Gifted Students to Work Together**
   According to NAGC, research shows that enabling gifted students to work together in groups boosts their academic achievement and benefits other students in the classroom, as well. When gifted students work together, they challenge themselves in unexpected ways. They bounce ideas off one another and take a peer's idea to a new place. They also learn that as smart as

they are, they, too, must exert effort with challenging content—and that they'll sometimes fail along the way.

6. **Plan for Tiered Learning**
   Author Carol Ann Tomlinson advocates *teaching up*—"a practice of first planning a lesson that's challenging for high-end learners and then differentiating for other learners by providing supports that enable them to access that more sophisticated learning opportunity." It replaces "the more common practice of planning for mid-range performers, then extending that lesson for advanced students and watering it down for others." This approach, Tomlinson says, challenges advanced learners more than trying to pump up a "middling" idea—and serves other students better as well.

There are several government programs implemented by the Department of Education and few private schools for the gifted in the Philippines.

1. Philippine High School for the Arts Model
2. Philippine Science High School System
3. Central Visayan Institute Foundation
4. The Learning Tree Child Growth Center
5. Headway School of Giftedness
6. BEREA Arts and Sciences High School

Note:

The Philippines is trying its best to provide gifted education programs for gifted students. In fact, in October 1999, then President Joseph Ejersito Estrada signed Presidential Proclamation No. 199 declaring every fourth week of November as the national week for the gifted and talented. Accordingly, one of the aims of the Philippine government in terms of education is to provide every gifted and talented Filipino with opportunities, encouragement, greater attention, and assistance to fully develop his or her potentials.

Application:

Quote the portion of the definition of giftedness by the authorities mentioned in this lesson under the following headings:

a. intelligence
b. creativity
c. talent
d. task commitment
e. leadership role

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
Organization	The introduction states the main topic and provides an overview of the essay. Information is relevant and presented in a logical order. The conclusion is good.	The introduction states the main topic and provides an overview of the essay. A conclusion is included.	The introduction states the main topic. A conclusion is included.	There is no clear introduction, structure, or conclusion.
Word Choice	It uses vivid words and phrases. The choice and placement of words seems accurate, natural, and not forced.	It uses vivid words and phrases. The choice and placement of words is inaccurate at times and/or seems overdone.	It uses words that communicate clearly, but the writing lacks variety.	It uses a limited vocabulary. Jargon or clichés may be present and detract from the meaning.
Sentence structure, grammar and mechanics	All sentences are well constructed and have varied structure and length. There are no errors in grammar, mechanics, and/or spelling.	Most sentences are well constructed and have varied structure and length. There are few errors in grammar, mechanics, and/or spelling, but they do not interfere with understanding.	Most sentences are well constructed, but they have a similar structure and/or length. There are several errors in grammar, mechanics, and/or spelling that interfere with understanding	Sentence's sound awkward, are distractingly repetitive, or are difficult to understand. There are numerous errors in grammar, mechanics, and/or spelling that interfere with understanding.

Adapted from: http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf

Summary of the Lesson:

The concept of giftedness varies among different scholars. It also varies from different cultures. Thus, there is no single definition of the word gifted that is universally accepted. Some scholars refer to it as having special talents and abilities while others think of it as a state of high intellect or genius.

The discussions here clearly indicate that giftedness and talent are complex condition that covers a wide range of human abilities and traits. As such, some students may excel in the academic subjects but may not show special talents in the arts. On the other hand, students who show outstanding talent in sports, visuals and arts, those with leadership abilities may show only average or above average performance in academic subjects.

Gifted and talented students need to be identified at an early age and be given support through the different assessments and programs within the schools and supported by the government.

Assessment:

I. Identification. Write the name of person or words that is being described in each statement.

______________________1. Considered as the highest expression of giftedness.

______________________2. Allows the students to engage in independent study, experimentation and investigation of topics that interest them.

________________________3. His definition of giftedness results from the interaction of above-average general abilities, high level of task commitment and creativity.
________________________4. According to him highly gifted students have IQ scores 3 SD or greater above the mean.
________________________5. He states that even if gifted students do not become producers of knowledge or makers of novelty, special education should train them to become adults who will produce knowledge.

II. True or False. Write True if the statement is correct and False if otherwise. Write the answer before each number.
________________________1. According to Gifted and Talented Children's Act of 1978, gifted and talented children are those who have potential abilities that give evidence of high performance capability.
________________________2. One of the identifiers of gifted and talented children according to Shaklee is being indifferent.
________________________3. Vertical enrichment is applied by expanding the content covered in Science, English and Filipino.
________________________4. Creative and insightful individuals are those who make discoveries and changes in the society.
________________________5. One of the characteristics of highly gifted individuals if perfectionism.

Enrichment Activity:

To know the future educators' perception about gifted and talented children, have an interview with 5 education students (could be your classmates or from other schools) about their own definition of giftedness and talent and come up with a thematic summary of their definitions.

Directions: Your essay will be graded based on this rubric. Consequently, use this rubric as guide in writing your essay and check it again before submitting your final output.

Criteria	5	3	2	1
Focus and Details	There is one clear, focused topic. Main idea is clear and well supported by detailed and accurate information.	There is one clear, well focused topic. Main idea is clear but not well supported by detailed information.	Main idea is somewhat clear.	The topic and main idea are not clear.
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Adapted from: *http://www.readwritethink.org/files/resources/printouts/Essay%20Rubric.pdf*

Suggested Links (Optional):

- link 1: https://study.com/academy/lesson/assessing-gifted-talented-students.html#lesson
- link 2 : https://www.rappler.com/brandrap/rich-media/72738-infographic-giftedness-talent

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- http://www.ascd.org/publications/newsletters/education-update/apr16/vol58/num04/Six-Strategies-for-Challenging-Gifted-Learners.aspx
- https://docs.education.gov.au/system/files/doc/pdf/gifted_talented_education_module6_primary.pdf